

Uncertainty-Aware Risk-Adjusted Benchmarking of the Fama–French Portfolios: Latent Performance, Rank Uncertainty, and the Limits of Forecastability

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Abstract

We benchmark the 25 Fama–French size and book-to-market portfolios on a validated balanced panel of 29,825 portfolio-months (1,193 months, July 1926–November 2025). The estimand is the Fama–French three-factor (FF3) alpha; the primary uncertainty object is the full 25×25 heteroskedasticity- and autocorrelation-consistent (HAC) covariance of the alpha vector; and the reported score is a multivariate empirical-Bayes posterior alpha. Rankings carry common-date block-bootstrap rank intervals and are evaluated out of sample. The joint HAC/Wald test rejects the hypothesis that all alphas are zero ($p = 6.4 \times 10^{-10}$), yet partial pooling, wide bootstrap rank intervals, and strong long-horizon mobility show that the point ranking is not a precise hierarchy. In strictly forward windows the top-minus-bottom quintile spread averages 243 bps/year. Our central contribution is an *incremental* test that decomposes this signal: framed as a Diebold–Mariano test of equal predictive accuracy, the time-varying rolling ranking adds no power over a leakage-free expanding-window ranking (mean skill differential -0.042 , HAC $t = -1.87$) and is dominated by a single fixed full-sample ordering. The forward result is therefore consistent with persistent characteristic mispricing, not dynamic forecasting skill. Alpha is benchmark-dependent (an 11-rank FF3-to-FF5 move), and only one of 25 stochastic-frontier boundary tests survives false-discovery control. The findings extend the skeptical and out-of-sample asset-pricing literatures to the question of whether *re-estimating* a cross-sectional ranking through time is worthwhile; here it is not.

Keywords: cross-sectional asset pricing; empirical Bayes; HAC inference; rank uncertainty; out-of-sample validation; predictive accuracy; false-discovery control.

JEL classification: G11, G12, C11, C58.

1 Introduction

A portfolio can earn a high average return because it bears systematic risk, because its sample happened to be favourable, because the benchmark is incomplete, or because it contains genuinely unusual performance. Factor benchmarking removes the part linearly associated with priced factors and studies the intercept, or alpha. For the 25 Fama–French portfolios—the canonical test assets

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from which the size and value factors are built (Fama and French 1993)—this intercept is a model-relative quantity: it is unexplained by the chosen factors, not by every conceivable risk or friction.

This paper makes the uncertainty around such a benchmark explicit and then asks a sharper question than the usual out-of-sample test. Our contributions are fourfold. First, we estimate the full cross-portfolio HAC covariance of the 25 alphas (Newey and West 1987) rather than 25 unrelated standard errors, and use it for the joint test of Gibbons, Ross, and Shanken (1989) and for multivariate empirical-Bayes shrinkage (Efron and Morris 1973). Second, we quantify ranking uncertainty with a common-date circular block bootstrap (Künsch 1989; Politis and Romano 1994) that refits every stage of the estimator. Third, we separate description from validation with a strictly forward evaluation in which scores, ranks, and factor loadings are frozen before the evaluation window. Fourth—and centrally—we introduce an *incremental* forward test that isolates whether the time-varying ranking adds anything beyond a fixed characteristic ordering. We show that this test is a Diebold–Mariano test of equal predictive accuracy (Diebold and Mariano 1995), with the Giacomini and White (2006) interpretation for rolling estimation schemes.

The headline results are as follows. The joint zero-alpha hypothesis is rejected under the dependence-robust HAC/Wald test, and five of 25 individual alphas survive 5% Benjamini–Hochberg false-discovery control (Benjamini and Hochberg 1995). Yet bootstrap rank intervals are wide and long-horizon quintile mobility is high, so the point ranking is not a reliable league table—an outcome anticipated by Lewellen, Nagel, and Shanken (2010). The strictly forward top-minus-bottom quintile spread is positive on average (243 bps/year). The incremental test then shows that this predictability is *not* dynamic: the 120-month rolling ranking beats a naive size/value characteristic gradient but does not beat a leakage-free expanding-window ranking, and is itself dominated by a single fixed full-sample ordering. The predictive content lives in a stable, irregular cross-sectional alpha pattern that FF3 misprices persistently, not in the re-estimation of the ranking through time. This mirrors, in the cross-section, the fragility of adaptive out-of-sample prediction documented by Welch and Goyal (2008) and the characteristic-based, decaying nature of anomaly returns in McLean and Pontiff (2016).

The remainder of the paper is organised as follows. Section 2 reviews the related literature. Section 3 describes the data and integrity checks. Section 4 sets out the estimator. Section 5 reports the empirical results, Section 6 the robustness and validation analysis, Section 7 discusses interpretation, and Section 8 concludes.

2 Related literature

Factor benchmarking and test assets. Fama and French (1993) introduced the three-factor model and the 25 size and book-to-market portfolios that remain the standard test assets, and Gibbons, Ross, and Shanken (1989) provided the finite-sample joint test that all intercepts are zero. Fama and French (2015) added profitability and investment factors, and Fama and French (2016) document that the model’s most serious pricing failures involve the smallest, lowest-book-to-market stocks—the “small-growth” cell. Cochrane (2011) reframes asset pricing around cross-sectional variation in expected returns. Our paper benchmarks exactly these test assets and reproduces the small-growth shortfall as the lowest-ranked cell.

Skepticism about alphas and cross-sectional tests. Lewellen, Nagel, and Shanken (2010) show that high cross-sectional fit and large apparent alphas on size/book-to-market portfolios can

be misleadingly strong, and recommend stricter standards, confidence intervals, and attention to the factor structure of the test assets. Harvey, Liu, and Zhu (2016) argue that the proliferation of candidate factors demands higher significance hurdles and multiple-testing corrections. These concerns motivate our use of the full joint HAC covariance, Benjamini–Hochberg false-discovery control, and bootstrap rank intervals rather than isolated t -statistics.

Shrinkage and empirical Bayes. The winner’s curse in ranking noisy estimates is classically addressed by Stein-type shrinkage and empirical Bayes (Efron and Morris 1973), applied to portfolio problems by Jorion (1986). Recent work shrinks the cross-section more aggressively—Kozak, Nagel, and Santosh (2020) on the pricing kernel and Gu, Kelly, and Xiu (2020) via machine learning—and finds that disciplined shrinkage and rich conditioning improve out-of-sample cross-sectional pricing. We use multivariate empirical Bayes with the full sampling covariance to temper extreme estimates; our negative dynamic result (below) concerns the time-variation of a coarse 25-cell ranking, not the value of conditioning information per se, so it does not contradict this literature.

Out-of-sample predictability and its fragility. Welch and Goyal (2008) show that many equity-premium predictors fail to beat the historical mean out of sample and that adaptively estimated models often underperform simple benchmarks. McLean and Pontiff (2016) find that anomaly returns are characteristic-based and decline substantially after publication. Our incremental finding—a stable ordering predicts future alpha while dynamic re-estimation does not—is a cross-sectional analogue of both: the signal is a persistent property of fixed characteristic cells, and chasing it with a rolling window adds estimation noise rather than information.

Forecast evaluation and bootstrap inference. Diebold and Mariano (1995) test equal predictive accuracy via the mean loss differential with HAC standard errors, and Giacomini and White (2006) extend this to models estimated on rolling windows. We cast the incremental test in exactly this framework. For finite-sample performance inference we follow the resampling tradition of Kosowski et al. (2006) and Fama and French (2010), using the block bootstrap of Künsch (1989) and Politis and Romano (1994) in a common-date, circular form; dependence-robust covariances follow Newey and West (1987) with the automatic bandwidth of Newey and West (1994). The secondary residual-asymmetry diagnostic uses the stochastic-frontier model of Aigner, Lovell, and Schmidt (1977) with the conditional-inefficiency estimator of Jondrow et al. (1982).

Contribution. Prior work either tests the alpha vector jointly (Gibbons, Ross, and Shanken 1989) or evaluates a cross-sectional ranking out of sample. We add a decomposition that separates the *dynamic* content of a re-estimated ranking from the *static* content of a fixed ordering, and show—using a Diebold–Mariano/Giacomini–White test on strictly forward data—that the dynamic component is statistically indistinguishable from zero for the 25 Fama–French portfolios.

3 Data

3.1 Portfolios and sample

The 25 test portfolios form a 5×5 grid. At each formation date eligible stocks are sorted independently into five size and five book-to-market groups; returns within each of the 25 intersections are

value weighted. Labels such as SMALL HIBM and BIG LOBM describe the two sorting dimensions, not individual firms or managed funds. The unit of observation is one portfolio-month; the dependent variable is the portfolio return minus the one-month risk-free rate, $y_{it} = r_{it} - r_{ft}$, in decimal monthly units. The canonical benchmark is FF3 (Mkt–RF, SMB, HML). Annualised basis points multiply monthly decimal alpha by $12 \times 10,000$, a linear reporting convention. The validated balanced panel has 29,825 portfolio-months over 1,193 consecutive months (July 1926–November 2025), with zero duplicate keys and no missing calendar months; the loader fails closed on any violation.

3.2 Integrity and provenance

Each run records the SHA-256 digest and byte count of both inputs. A current (May 2026) official snapshot from the Kenneth R. French Data Library is retained separately and never overwrites the fixed local inputs. The local sample’s exact acquisition vintage is unverified, and the two sources differ over their overlap—by up to 1.4167 percentage points for portfolio returns and 0.34 percentage points for FF3/RF fields. Official historical revisions are a plausible but unproven cause. Both comparisons are therefore classified NOT_COMPARABLE (visible, non-blocking) rather than PASS. This is a genuine limitation: the headline numbers rest on inputs that cannot currently be reconstructed byte-for-byte from the public source, even though all within-pipeline integrity and independent-replication checks pass (Section 6).

4 Methodology

4.1 Factor model and the meaning of alpha

For portfolio i in month t , the benchmark regression is

$$y_{it} = \alpha_i + \beta_i' f_t + \varepsilon_{it}, \quad (1)$$

where f_t stacks the FF3 factor returns and α_i is the mean excess return left after the fitted linear factor component is removed. Ordinary least squares gives $\hat{\theta}_i = (X'X)^{-1}X'y_i$; selecting the intercept, the estimation error is a weighted sum of residual shocks, $\hat{\alpha}_i - \alpha_i = \sum_t a_t \varepsilon_{it}$, where the scalar weight a_t depends only on the common factor design. Because every portfolio shares the same dates and factors, these weights combine with the same-month residual vector to yield a *joint* covariance. A positive alpha denotes in-sample benchmark-relative outperformance, not skill or future profit; a negative alpha can reflect omitted risk rather than inefficiency.

4.2 Full joint HAC covariance

Let e_t be the 25-vector of same-month residuals and $g_t = a_t e_t$ the alpha score vector. With $L = 12$ lags, Bartlett weights $w_l = 1 - l/(L + 1)$, and lag- l cross-product $\Gamma_l = \sum_{t>l} g_t g_{t-l}'$, the covariance is

$$V_{\text{HAC}} = \frac{T}{T - P} \left\{ \Gamma_0 + \sum_{l=1}^L w_l (\Gamma_l + \Gamma_l') \right\}, \quad (2)$$

with $P = K + 1$ regression coefficients (Newey and West 1987). This 25×25 matrix keeps individual alpha variances on its diagonal and cross-portfolio co-movement off-diagonal. The implementation symmetrises the result, checks eigenvalues, and permits only scale-relative flooring of negligible numerical negatives. The canonical matrix is positive definite without flooring, with condition number 139.4.

4.3 Multivariate empirical Bayes

Ranking 25 noisy estimates invites a winner’s curse. Partial pooling treats the latent alphas as one cross-section,

$$\hat{\alpha} \mid \alpha \sim N(\alpha, V_{\text{HAC}}), \quad \alpha \sim N(\mu \mathbf{1}, \tau^2 I), \quad (3)$$

with posterior mean $\mu \mathbf{1} + \tau^2(V_{\text{HAC}} + \tau^2 I)^{-1}(\hat{\alpha} - \mu \mathbf{1})$ and conditional covariance $\tau^2 I - \tau^4(V_{\text{HAC}} + \tau^2 I)^{-1}$. The hyperparameters (μ, τ) are estimated by profile marginal maximum likelihood, so the prior is learned from the same cross-section (Efron and Morris 1973; Jorion 1986). The fitted prior mean is -26 bps/year with dispersion 94 bps/year. Reported intervals include first-order uncertainty in the profiled mean but do not fully integrate uncertainty in τ , and the HAC covariance is treated as known; the intervals are therefore mildly optimistic model-based summaries.

4.4 Joint and multiple testing

The primary joint test is the dependence-robust Wald statistic $W = \hat{\alpha}' V_{\text{HAC}}^{-1} \hat{\alpha} \sim \chi^2(25)$ under the null that all alphas are zero. The Gibbons, Ross, and Shanken (1989) F -test is reported as a conventional secondary benchmark; its exact reference assumes IID Gaussian errors, which the diagnostics reject (Section 6). Individual raw-alpha p -values and the 25 stochastic-frontier boundary p -values form two separate Benjamini–Hochberg families (Benjamini and Hochberg 1995), since they test different nulls.

4.5 Bootstrap rank uncertainty

A rank is discontinuous, so individual standard errors do not describe how often a portfolio stays in the top five. We draw 5,000 circular 12-month blocks of common date indices (Künsch 1989; Politis and Romano 1994), apply the same dates to all 25 portfolios (preserving cross-sectional shocks), and refit the regressions, the full HAC covariance, and the empirical-Bayes hyperparameters within every replicate before re-ranking. Rank and score intervals are empirical 2.5–97.5% quantiles of this resampling distribution.

4.6 Forward validation and the incremental test

For each 120-month training window we freeze all scores, ranks, prior parameters, and betas at the cutoff, then compute each portfolio’s next 12-month benchmark-adjusted return with the frozen loadings, $A_i^+ = y_{it} - \hat{\beta}_i' f_t$. Within each evaluation year we compute the Spearman correlation between training scores and future alpha and the average top-minus-bottom quintile spread; annual statistics are aggregated with a Fisher transformation and four-lag HAC inference. Evaluation windows do not overlap.

The forward test alone cannot separate genuine dynamic skill from persistent characteristic mispricing: because FF3 misprices fixed characteristic cells in a stable way, a ranking anchored to past alpha will correlate with future alpha mechanically. We therefore add an *incremental* test that scores the rolling ranking against a competing benchmark ordering on the *same* forward observations. For window j let d_j be the difference between the rolling ranking’s predictive score (Spearman correlation with future alpha, or the quintile spread) and the benchmark’s. The mean differential $\bar{d}$ with a HAC standard error is exactly the Diebold and Mariano (1995) test of equal predictive accuracy, and because the competing rankings are re-estimated on rolling/expanding

schemes it carries the Giacomini and White (2006) interpretation. We use four benchmark orderings: (i) a size-ascending, book-to-market-descending characteristic gradient; (ii) a diagonal value-minus-size gradient; (iii) a leakage-free *expanding-window* ranking re-estimated on all data up to each cutoff; and (iv) a full-sample *oracle* ordering, which uses look-ahead information and is thus an upper bound on what a static ranking can achieve. Benchmark (iii) is the fair, tradable comparator on which the interpretation rests. The per-window differential is aggregated with an automatic-bandwidth Newey–West HAC *t*-test (Newey and West 1994) and a block bootstrap.

5 Results

5.1 Cross-sectional posterior performance

Table 1 reports the full ranking. SMALL HIBM has the highest posterior alpha (132 bps/year) and SMALL LOBM the lowest (−185 bps/year), reproducing the well-known small-value premium and small-growth shortfall relative to FF3 (Fama and French 2016). Two posterior intervals are wholly above zero (SMALL HIBM, BIG LOBM); most cross zero. Five raw-alpha tests survive Benjamini–Hochberg control, but four of these correspond to significantly negative alpha, so raw-test significance and posterior rank are distinct summaries.

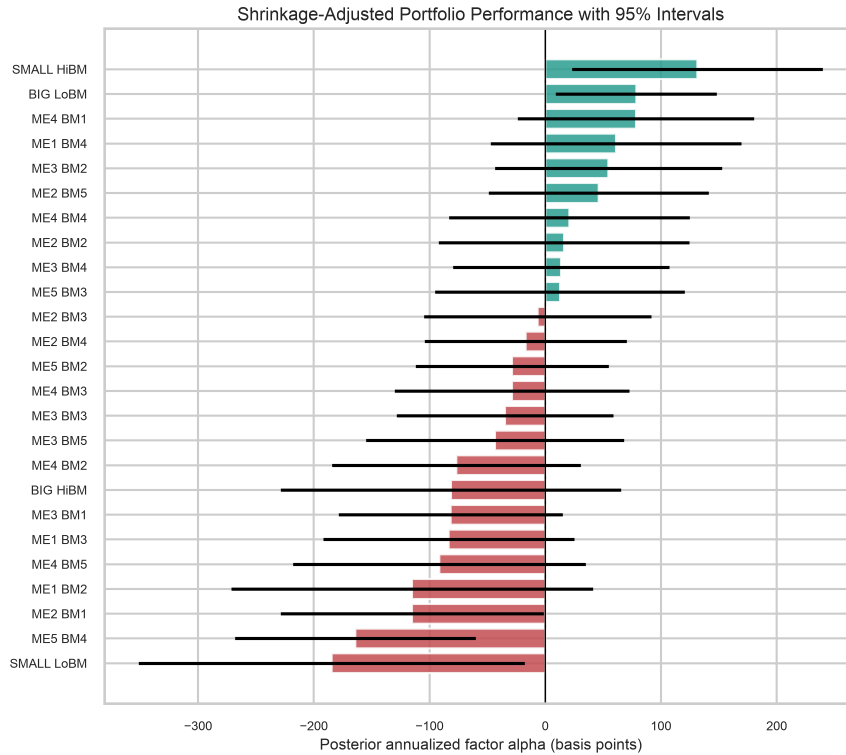


Figure 1: Posterior FF3 alpha (annualised bps) with approximate 95% posterior intervals. An interval crossing zero indicates the latent alpha is not cleanly separated from zero; interval overlap warns that adjacent ranks are not distinguished.

Table 1: Posterior FF3 performance ranking (annualised basis points). Raw and posterior alpha in bps/year; posterior 95% is the empirical-Bayes interval; BH q is the false-discovery-adjusted raw-alpha p -value; the bootstrap rank interval is the central 95% range over 5,000 common-date circular-block draws.

Rank	Portfolio	Raw α	Post. α	Post. 95%	BH q	Boot. rank 95%
1	SMALL HiBM	169	132	[23, 240]	0.063	[1.0, 6.0]
2	BIG LoBM	118	79	[9, 148]	0.033	[1.0, 10.0]
3	ME4 BM1	50	78	[-24, 180]	0.732	[1.0, 15.0]
4	ME1 BM4	84	61	[-47, 169]	0.456	[1.0, 12.0]
5	ME3 BM2	103	55	[-43, 153]	0.227	[1.0, 12.0]
6	ME2 BM5	37	46	[-49, 141]	0.766	[2.0, 15.0]
7	ME4 BM4	23	21	[-83, 125]	0.861	[2.0, 18.0]
8	ME2 BM2	21	16	[-92, 125]	0.868	[2.0, 18.0]
9	ME3 BM4	44	14	[-80, 107]	0.732	[3.0, 18.0]
10	ME5 BM3	-38	13	[-95, 121]	0.825	[3.0, 19.0]
11	ME2 BM3	26	-7	[-105, 92]	0.825	[3.0, 19.0]
12	ME2 BM4	25	-17	[-104, 70]	0.825	[4.0, 19.0]
13	ME5 BM2	-5	-29	[-112, 55]	0.912	[5.0, 20.0]
14	ME4 BM3	-15	-29	[-130, 73]	0.874	[6.0, 21.0]
15	ME3 BM3	52	-35	[-128, 59]	0.732	[4.0, 20.0]
16	ME3 BM5	-76	-43	[-155, 68]	0.562	[6.0, 22.0]
17	ME4 BM2	-14	-77	[-184, 31]	0.874	[9.0, 23.0]
18	BIG HiBM	-198	-82	[-229, 65]	0.336	[10.0, 24.0]
19	ME3 BM1	-142	-82	[-178, 15]	0.055	[11.0, 23.0]
20	ME1 BM3	-144	-83	[-192, 25]	0.106	[9.0, 23.0]
21	ME4 BM5	-211	-92	[-218, 35]	0.063	[13.0, 24.0]
22	ME1 BM2	-476	-115	[-271, 41]	0.005	[15.0, 25.0]
23	ME2 BM1	-261	-115	[-229, -1]	0.003	[15.0, 25.0]
24	ME5 BM4	-289	-164	[-268, -60]	0.002	[19.0, 25.0]
25	SMALL LoBM	-810	-185	[-351, -18]	<0.001	[21.0, 25.0]

5.2 Joint tests and rank uncertainty

The HAC/Wald statistic is 93.9 ($p = 6.4 \times 10^{-10}$) and the GRS statistic is 3.28 ($p = 1.2 \times 10^{-7}$); both reject the joint zero-alpha hypothesis, though the GRS reference is less credible here. Rejection means the 25-dimensional alpha vector is inconsistent with all-zero under the stated model; it does not establish that alphas are positive or that the ordering is stable. Figure 2 shows the bootstrap rank distribution: SMALL HIBM is rank 1 at the point estimate and in the top quintile in a majority of resampled histories, yet its central rank interval still reaches rank 6, and intervals overlap broadly across portfolios. Consistent with Lewellen, Nagel, and Shanken (2010), the ordered list is not a precise hierarchy.

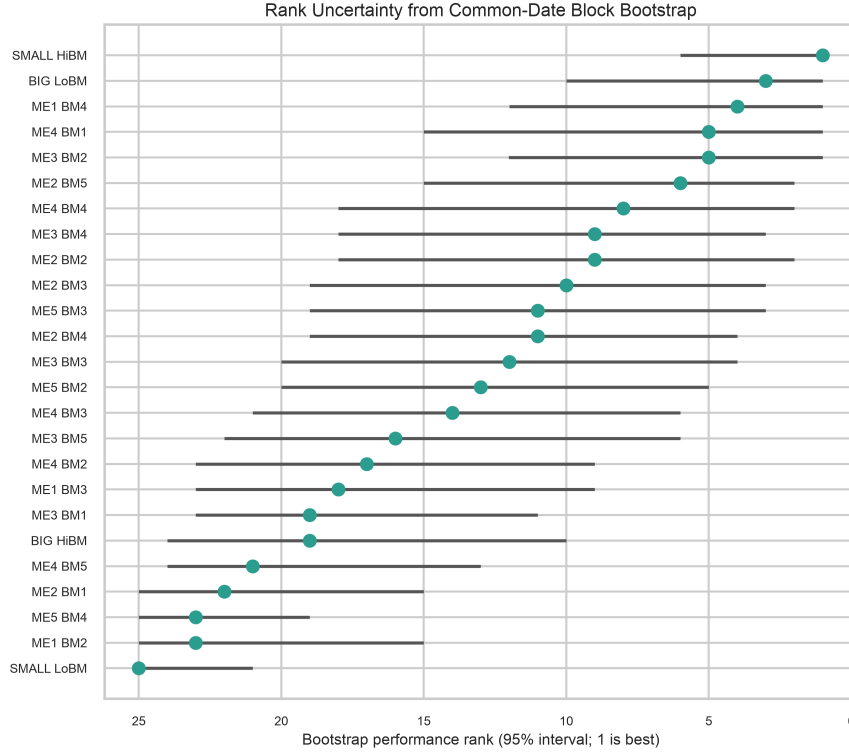


Figure 2: Median bootstrap rank and central 95% rank interval from 5,000 common-date circular-block draws. Broad, overlapping intervals are direct evidence against reading the ordered list as a precise hierarchy.

5.3 Persistence

Rank persistence falls from about 0.86 at a 12-month horizon (90% window overlap) to roughly 0.16 at 120 months (zero overlap). The non-overlapping horizon is the credible structural measure and is weak: the mean probability of staying in the same quintile after 120 months is 24.9%, essentially the 20% implied by uniform mobility. Long-run relative performance is highly mobile.

5.4 Strictly forward validation

Across 89 non-overlapping 12-month evaluation periods, the Fisher-averaged rank correlation between training scores and future alpha is 0.102 (HAC 95% CI [0.055, 0.149], $p = 2.3 \times 10^{-5}$), and the average top-minus-bottom quintile spread is 243 bps/year (median 209, IQR 498), positive in 73% of windows (Table 2). Both means are statistically distinguishable from zero, but dispersion across years is large—the IQR exceeds the mean—so the average is not a uniformly reliable timing rule (Figure 3).

Table 2: Strictly forward validation (four-lag HAC inference), 89 non-overlapping 12-month windows.

Metric	Estimate	HAC 95% interval	HAC p
Rank vs. future alpha (Spearman)	0.102	[0.055, 0.149]	2.3×10^{-5}
Top–bottom quintile (bps/year)	243	[153, 333]	1.3×10^{-7}

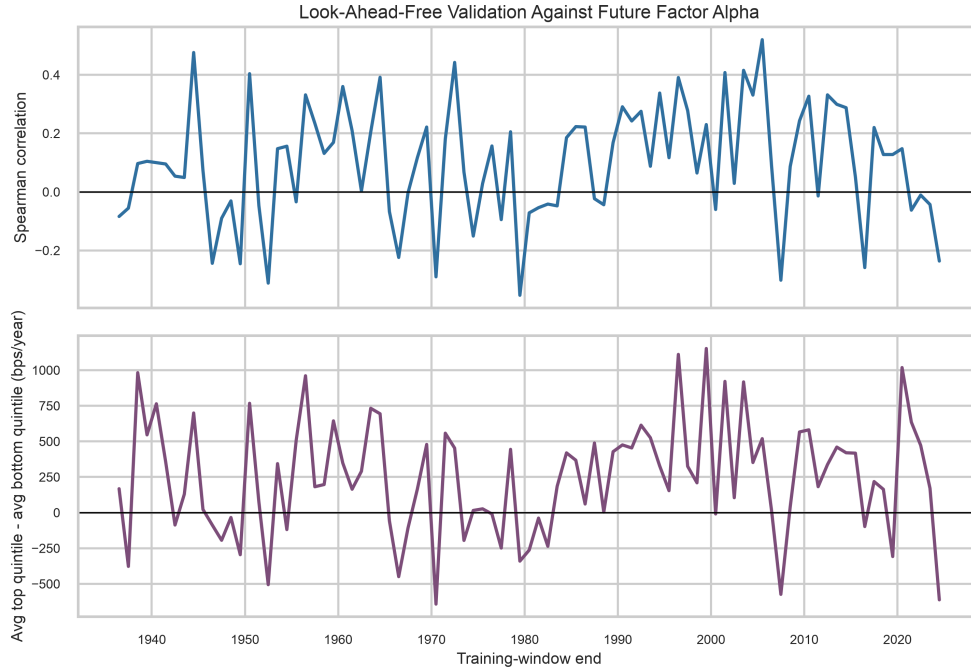


Figure 3: Window-by-window future rank correlation (top) and average top-minus-bottom quintile future alpha (bottom). Frequent negative windows and large spread variation show the average is not uniformly reliable.

5.5 The incremental test: is the ranking dynamically informative?

Table 3 decomposes the forward signal. The rolling ranking’s own level correlation with future alpha is +0.099. It exceeds the two static characteristic gradients—which have essentially zero forward power, because FF3 alpha is the residual after size and value exposure are removed—so the rolling signal is not merely a restated size/value tilt. But, read as a Diebold–Mariano test, it does not

beat the leakage-free expanding-window ranking: the mean skill differential is -0.042 with HAC $t = -1.87$ (the block bootstrap 95% interval $[-0.085, +0.001]$ includes zero), so we cannot reject equal predictive accuracy, and the point estimate marginally favours the expanding benchmark. Against the full-sample oracle the differential is significantly negative (-0.100 , $t = -4.18$). A single fixed but irregular cross-sectional ordering predicts future alpha as well as, or better than, the ranking that is re-estimated every year (Figure 4).

Table 3: Incremental predictive power of the rolling ranking versus fixed and leakage-free benchmarks (Diebold–Mariano equal-predictive-accuracy tests on 89 forward windows; automatic-bandwidth HAC, lag 3). “Bench. corr” is the benchmark ordering’s own mean Spearman correlation with future alpha; the rolling ranking’s own level is $+0.099$. Δ rank corr is the mean per-window skill differential (rolling minus benchmark); t is its HAC t -statistic; p is the one-sided test that the differential exceeds zero; Δ spread is the mean top-minus-bottom spread differential (bps/year). The oracle uses look-ahead information and is an upper bound, not a tradable strategy.

Benchmark ordering	Bench. corr	Δ rank corr	t	p	Bootstrap 95% CI	Δ spread
Characteristic gradient (size $\uparrow$, B/M $\downarrow$)	-0.018	$+0.116$	$+4.36$	<0.001	$[+0.062, +0.169]$	$+343$
Characteristic gradient (diagonal)	$+0.006$	$+0.092$	$+3.39$	<0.001	$[+0.038, +0.145]$	$+221$
Expanding window (leakage-free)	$+0.140$	-0.042	-1.87	0.969	$[-0.085, +0.001]$	-44
Full-sample alpha (oracle)	$+0.199$	-0.100	-4.18	1.000	$[-0.148, -0.053]$	-165

The interpretation is unambiguous and consistent with Welch and Goyal (2008) and McLean and Pontiff (2016): the out-of-sample predictability documented above is real but reflects a stable, irregular pattern of characteristic mispricing that a fixed ranking captures at least as well; discarding data older than the rolling window and chasing recent alpha adds estimation noise, not forecasting skill. Claims of *dynamic* forecasting ability are not supported by the data.

6 Robustness and validation

Benchmark sensitivity (FF3 vs. FF5). On the prespecified common sample (July 1963–November 2025), FF3 and FF5 posterior ranks have Spearman correlation 0.875 . ME5 BM2 moves from rank 9 under FF3 to rank 20 under FF5, an 11-position change; FDR-supported raw alphas move from five to three; and HAC/Wald rejects joint zero alpha under both models. Following Fama and French (2015), part of what looks like FF3 performance is exposure to profitability and investment dimensions absent from FF3, confirming that alpha is benchmark-dependent.

Window-length sensitivity. Rank correlations across 60-, 120-, and 180-month windows are positive but far from one, with top-quintile membership overlap (Jaccard) that can fall below 0.40 . Window length is not a neutral tuning choice; disagreement reflects genuine temporal instability.

Residual diagnostics. Residual normality is rejected for all 25 portfolios (Jarque–Bera), serial independence for 16 of 25 (Ljung–Box, lag 12), and conditional homoskedasticity for all 25 (ARCH-LM, lag 12). An IID-Gaussian error model is not credible for these returns, which is why we rely on HAC inference and block resampling and treat GRS as secondary. On the historical 1963–1991 anchor, HAC/Wald rejects while GRS does not, consistent with this diagnostic evidence.

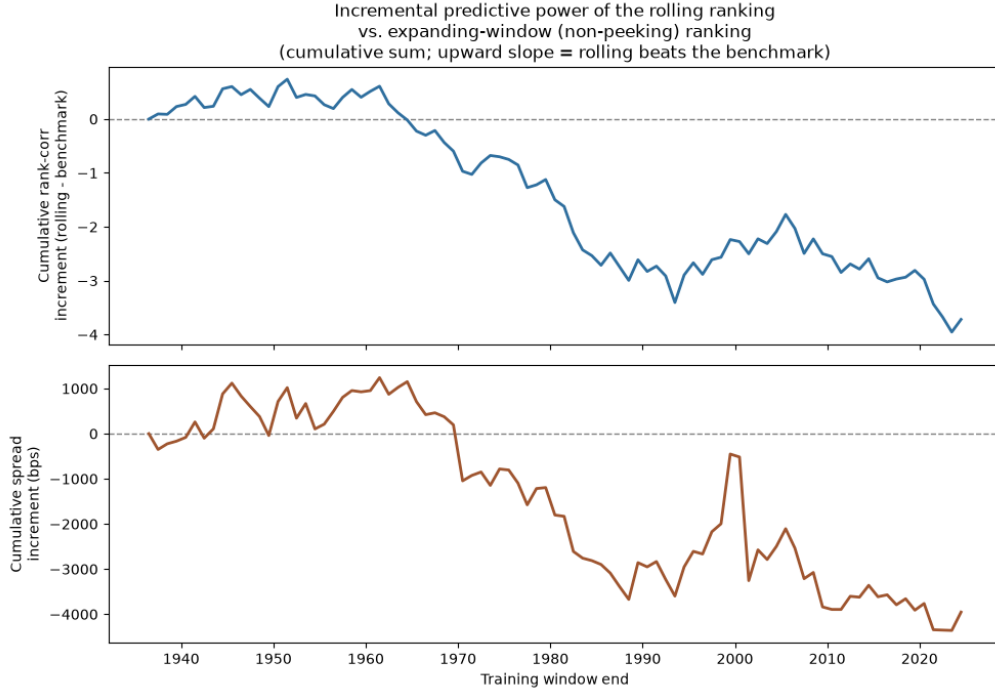


Figure 4: Cumulative rolling-minus-expanding-window differential in rank correlation (top) and quintile spread (bottom). A flat or downward path shows the time-varying ranking does not out-predict a ranking that simply accumulates all past data.

Independent replication. All coefficients and HAC standard errors reproduce independently with `statsmodels` to a tolerance of 10^{-12} for all 25 portfolios, and the qualitative 5×5 patterns, excess-return identity, block selection, and weighting checks pass. Independent numerical replication is distinct from source reconciliation: the two current-snapshot comparisons remain NOT_COMPARABLE (Section 3) because the local acquisition vintage is unverified.

Stochastic-frontier diagnostic. A half-normal stochastic-frontier model (Aigner, Lovell, and Schmidt 1977) with the conditional-inefficiency estimator of Jondrow et al. (1982) is applied only as a residual-asymmetry diagnostic; a free intercept and a one-sided shortfall are not separately identified, so it does not substitute for α . The boundary null $\sigma_u = 0$ is tested with the appropriate 50:50 χ^2 mixture and Benjamini–Hochberg control. Only one of 25 boundary tests survives 5% FDR, so no general efficiency ranking is warranted.

7 Discussion

The central message is a distinction the raw forward test obscures: a *stable* cross-sectional ordering of the 25 cells carries out-of-sample information, but the *time variation* of the rolling ranking does not. This sits naturally within the skeptical asset-pricing literature. The wide rank intervals and the 11-position FF3-to-FF5 move echo Lewellen, Nagel, and Shanken (2010) and Harvey, Liu, and Zhu (2016); the fragility of the adaptive scheme relative to a fixed ordering parallels Welch and Goyal (2008); and the characteristic-based, persistent nature of the predictable component

matches McLean and Pontiff (2016) and the cross-sectional emphasis of Cochrane (2011). Our result does not contradict the finding that disciplined shrinkage and rich conditioning improve cross-sectional pricing (Kozak, Nagel, and Santosh 2020; Gu, Kelly, and Xiu 2020): those gains come from conditioning information and regularisation, whereas we study whether re-estimating a coarse 25-cell ranking through time adds value—and it does not.

The most defensible use of the framework is comparative and uncertainty-aware: estimate benchmark-relative performance, shrink noisy extremes, and report rank uncertainty. The least defensible use is to select a single portfolio from the point rank, or to treat the rolling ranking’s apparent timing ability, as a guaranteed future excess return.

Limitations. Returns exclude transaction costs, taxes, shorting frictions, capacity, and market impact; the top-minus-bottom construction shorts small-growth, which is costly and capacity-constrained in practice. The source portfolios are value-weighted research constructions, not live investable funds, and full-sample alphas embed later data revisions. The factor model is linear with constant betas within each window. Posterior intervals do not fully integrate hyperparameter uncertainty and treat the HAC covariance as known, so they are mildly optimistic. The oracle benchmark uses look-ahead information and is an upper bound; the expanding-window benchmark is the leakage-free comparator on which the negative incremental verdict rests. Finally, the local input vintage cannot currently be reconciled byte-for-byte with the public source.

8 Conclusion

This paper provides an uncertainty-aware benchmark for the 25 Fama–French portfolios built on a validated panel, a full joint HAC covariance, multivariate empirical-Bayes shrinkage, multiplicity control, overlap-aware persistence analysis, and strictly forward validation. Cross-sectional FF3 alpha variation is present and jointly significant, but its apparent precision is substantially reduced by partial pooling, bootstrap rank uncertainty, temporal mobility, and benchmark sensitivity. The main contribution is the incremental forward test: framed as a Diebold–Mariano/Giacomini–White comparison, it shows that the positive out-of-sample quintile spread is driven by persistent characteristic mispricing captured by a stable ordering, not by dynamic skill in the re-estimated ranking. Future work should test whether any residual signal survives realistic turnover and trading costs, whether additional prespecified factors absorb the FF3 residual alphas without specification search, and whether a decision-theoretic allocation over rank probabilities improves on hard top-quintile selection. Any real-time claim should also await reconciliation of the data vintage with the official source.

A Reproducibility

The reviewed environment is CPython 3.13; the deterministic bootstrap uses seed 2026. Source hashes, settings, and runtime are persisted in `results/run_manifest.json`. The full pipeline is `pip install -r requirements-dev-lock.txt; pytest -q; ruff check .; MPLBACKEND=Agg python -m analysis.run_all; python -m analysis.export_tables; python reports/generate_report.py`. The incremental test is produced within `analysis.run_all` and can be regenerated with `python -m analysis.incremental_validation -scheme all`.

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